CLASS 11-CBSE MATHEMATICS

CHAPTER: SEQUENCES AND SERIES-SET-1

TIME : 1Hr **Mx.Marks:30**

SECTION A

Q.I Answer the following questions **[1X3 = 3 MARKS]**

1. Let T_r be rth term of an A.P. whose first term is a and common difference is d. If for some positive integers $m, n, m \neq n$, $T_m = \frac{1}{n}$ and $T_n = \frac{1}{m}$, then find $a - d$
2. If the 9th term of an A.P. be zero, then find the ratio of its 29th and 19th term.
3. If 7th and 13th term of an A.P. be 34 and 64 respectively, then find its 18th term.

SECTION B

Q.II Answer the following questions **[2X5 = 10 MARKS]**

4. For an A.P., $T_2 + T_5 - T_3 = 10,\quad T_2 + T_9 = 17,$ then common difference is
5. Which term of the sequence (– 8 + 18i), (– 6+15i), (– 4 + 12i), ...is purely imaginary.
6. If the sum of first p terms, first q terms and first r terms of an A.P. be x, y and z respectively, then Prove that
$$\frac{x}{p}(q-r)+\frac{y}{q}(r-p)+\frac{z}{r}(p-q)=0$$
.
7. If $x, 2x+2, 3x+3$ are in G.P., then find the fourth term.
8. Find the sum of odd integers from 1 to 2001.

SECTION C

Q.III Answer the following questions **[3X3 = 9 MARKS]**

9. If the sum of first p terms of an A.P. is equal to the sum of the first q terms, then find the sum of the first (p + q) terms.
10. The ratio of sum of m and n terms of an A.P. is $m^2 : n^2$, then find the ratio of mth and nth term.

11. Show that the sum of all odd integers between 1 and 1000 which are divisible by 3 is 83667.

SECTION D

Q.IV Answer the following questions **[4X2 = 8 MARKS]**

12. Shamshad Ali buys a scooter for Rs 22000. He pays Rs 4000 cash and agrees to pay the balance in annual installment of Rs 1000 plus 10% interest on the unpaid amount. How much will the scooter cost him?

13. A farmer buys a used tractor for Rs 12000. He pays Rs 6000 cash and agrees to pay the balance in annual installments of Rs 500 plus 12% interest on the unpaid amount. How much will be the tractor cost him?